

Prince Sattam bin Abdulaziz University

Master of Data Science Program

Large-Scale Arabic Sentiment Analysis on YouTube Comments: LLM-Assisted Labeling and Transformer Fine-Tuning

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Text Mining IS 0661 – Course Project

GitHub Repository:

<https://github.com/AbdulrhmanFahd/text-minig-project>

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Abstract

In this paper, I present a comprehensive, end-to-end Sentiment Analysis (SA) pipeline specifically designed for complex Arabic content. Processing Arabic text poses unique challenges due to its morphological richness, dialectal variations, and orthographic inconsistencies, particularly within unstructured social media data. The dataset utilized in this research was extracted entirely from YouTube comments, representing a diverse array of Arabic dialects and nationalities. To address the chronic scarcity of large-scale labeled datasets in Arabic Natural Language Processing (NLP), I applied an innovative hybrid approach for text mining. By utilizing the Batch API service in Google AI Studio, I successfully automated the annotation of approximately 138,000 YouTube comments using Gemini-2.5 Flash. The primary outcome of this text mining effort was the construction of a robust, large-scale dataset that enabled the training and fine-tuning of a specialized transformer model (CAMELBERT). The developed model demonstrates high proficiency in predicting binary sentiment from highly contextual Arabic comments. I outline the architectural evolution of my methodology, comparing contemporary Deep Learning techniques against traditional Machine Learning baselines such as Term Frequency-Inverse Document Frequency (TF-IDF) combined with Support Vector Machines (SVM) and Logistic Regression (LR). The final evaluation reveals that while traditional models overfit the majority class, the proposed transformer-based approach demonstrates superior true semantic understanding, achieving a robust Macro-F1 score of 72.83% and an Accuracy of 76.38%.

Keywords: Arabic Sentiment Analysis, LLM Labeling, Transformers, Dialectal Arabic, Text Mining, Deep Learning, Natural Language Processing.

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1. Introduction and Problem Definition

In the modern digital age characterized by rapid technological advancement, vast amounts of text data are generated daily across various social media platforms. YouTube, in particular, serves as a massive repository of user-generated content where individuals express their opinions, feedback, and sentiments in the comment sections. For the Arabic-speaking world, this text data is a goldmine of public opinion. However, the sheer volume of this data makes it virtually impossible to monitor, guard, or analyze manually.

To address this information overload, Artificial Intelligence (AI) and Natural Language Processing (NLP) provide vital solutions. By applying text mining methodologies, we can build a Sentiment Analysis (SA) model designed to automatically classify the sentiment of a given text as either positive or negative. While the theoretical solution is straightforward, the practical implementation for the Arabic language is fraught with challenges. The primary obstacle is the acquisition of large, high-quality, labeled datasets required to train robust deep learning models.

Furthermore, Arabic NLP presents complex linguistic hurdles. The language is characterized by high ambiguity, heavy reliance on context, diverse morphological structures, and a severe lack of diacritics in informal writing. Additionally, the dichotomy between Modern Standard Arabic (MSA) and the multitude of regional dialects (e.g., Saudi, Egyptian, Levantine) complicates the analysis. A word or syntactic structure that denotes a positive sentiment in the Saudi dialect might carry a completely different, or even negative, connotation in the Egyptian dialect. Therefore, relying on advanced, context-aware AI models is not just an option, but an absolute necessity to accurately capture the semantic nuances of dialectal Arabic.

2. Literature Review

The evolution of Arabic Sentiment Analysis has transitioned through various phases, from early lexicon-based approaches to traditional machine learning, and recently, to advanced deep learning and transformer-based architectures. Recent academic studies have heavily underscored the efficacy of transformers in tackling the complexities of ASA.

Foundational research by Alakrot et al. [Alakrot et al., 2018] laid the critical groundwork for dataset construction in Arabic social media. Their work highlighted the difficulties of collecting and manually annotating antisocial behavior and general sentiments in YouTube comments, emphasizing the severe bottleneck that manual annotation creates in Arabic NLP research.

More recently, comprehensive surveys such as the one conducted by Almurqren et al. [Almurqren et al., 2025] have meticulously mapped the contemporary landscape of ASA chal-

lenges. Their survey points out that while MSA is relatively well-resourced, dialectal Arabic (DA) remains a significant challenge due to the lack of standardized spelling and the continuous evolution of internet slang.

To combat these linguistic variations, recent advancements have focused on hybrid deep learning architectures. Alosaimi et al. [Alosaimi et al., 2024] demonstrated that architectures combining pre-trained transformer models like AraBERT with recurrent neural networks (such as Long Short-Term Memory networks or BiLSTMs) significantly improve the model’s ability to capture dialectal context and long-term temporal dependencies in long comments. Similarly, Ouza et al. [Ouza et al., 2024] showed that fine-tuning AraBERT on specific dialectal corpora yields vastly superior classification metrics compared to traditional frequency-based models.

Furthermore, emerging literature is beginning to explore the role of Large Language Models (LLMs) in the NLP pipeline. Zouidine and Khalil [Zouidine and Khalil, 2025] highlighted the emerging role of LLMs not just as conversational agents, but as powerful tools for complex zero-shot and few-shot classification tasks in Arabic.

This project bridges the current research gap by leveraging the power of LLMs not merely as the final classification mechanism, but as a high-throughput, automated text mining tool for large-scale dataset construction. By solving the annotation bottleneck using LLMs, we pave the way for efficient and highly accurate transformer fine-tuning.

3. Dataset Acquisition and Exploratory Data Analysis

3.1 Data Consolidation

The foundation of any robust NLP model is its data. The raw data utilized in this research originated from a collection of 150 separate Excel files containing YouTube comments, initially prepared during Alakrot’s research. These comments were scraped from diverse Arabic YouTube channels covering various topics. In total, the raw collection comprised approximately 130,000 to 150,000 comments. However, only a fraction of these (roughly 16,000) were manually labeled, which is highly insufficient for training a modern deep learning transformer model without severe overfitting. The first critical step was consolidating these 150 disparate files into a single, unified dataset to prepare it for large-scale automated labeling.

3.2 Exploratory Data Analysis (EDA)

Before feeding the data into any labeling or training pipeline, it is imperative to understand the underlying structure, distribution, and linguistic characteristics of the corpus. I conducted an extensive Exploratory Data Analysis (EDA) to visualize these dimensions.

As shown in Figure 1, the dataset is not monolithic; it contains a vibrant mixture of regional

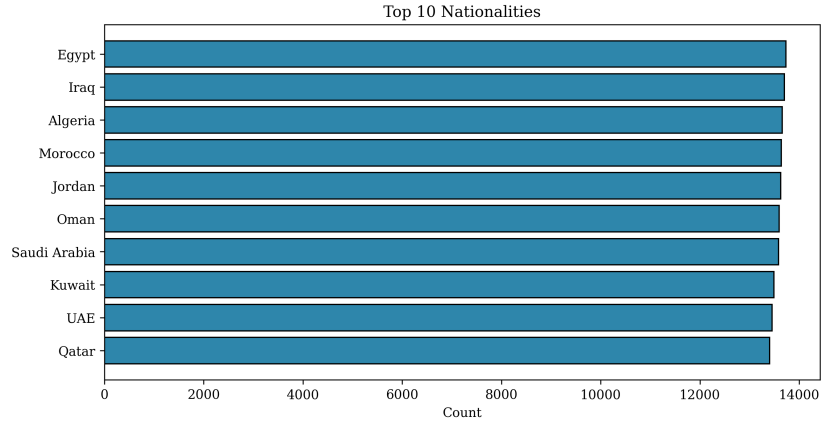


Figure 1: Distribution of Dialects and Nationalities in the YouTube Comments

dialects. Understanding the distribution of these nationalities is critical because it justifies the selection of the underlying language model. The presence of Gulf, Egyptian, and Levantine dialects necessitates the use of a model like CAMELBERT-Mix, which is explicitly pre-trained on a diverse corpus of varied Arabic dialects rather than just formal news articles.

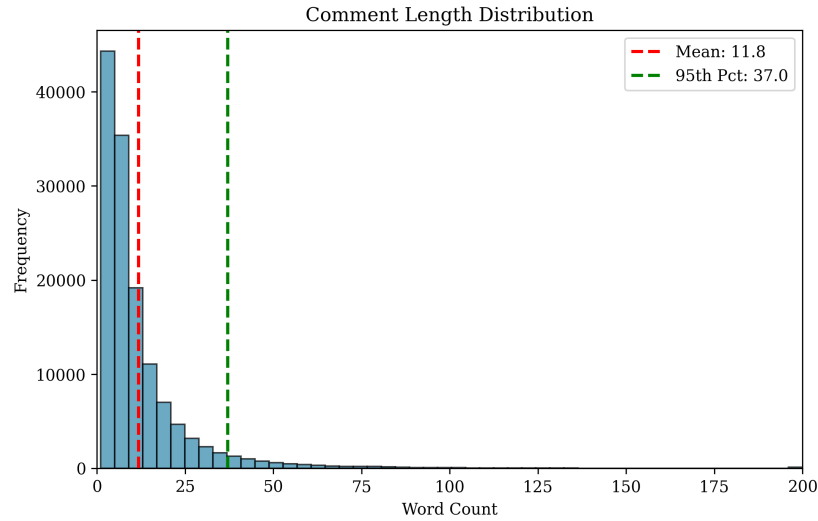


Figure 2: Distribution of Comment Lengths (Word Count)

Figure 2 illustrates the distribution of comment lengths across the dataset. The x-axis represents the number of words, while the y-axis shows the frequency. Most comments are relatively short, concise bursts of text (between 5 to 20 words). Analyzing this length distribution is crucial for setting the `'max_length' parameter in the transformer tokenizer to optimize memory usage during training`.

4 4. LLM-Assisted Text Mining and Labeling Pipeline

4.1 4.1 The Annotation Bottleneck

After consolidating the dataset, the primary obstacle was the lack of labels. Manual annotation of over 130,000 comments is prohibitively expensive and time-consuming, often requiring months of human labor and rigorous inter-annotator agreement protocols. To overcome this, I engineered an automated text mining pipeline.

4.2 4.2 Gemini-2.5 Flash Batch Processing

The solution leveraged one of the most advanced LLMs available for the Arabic language: Gemini-2.5 Flash. Using the Google AI Studio Batch API, I orchestrated a massive automated labeling campaign. The process involved sending approximately 10,000 automated requests, where each request batched 5 comments to optimize throughput and reduce API overhead.

I utilized a "few-shot" prompting strategy, providing the LLM with strict instructions and examples of how to classify dialectal Arabic into binary categories (Positive = 1, Negative = 0). The entire processing of over 130,000 comments was completed in merely 3 hours.

Crucially, I prompted the LLM to output not just the binary label, but also a mathematical **confidence score** ranging from 0.0 to 1.0. To maintain strict dataset integrity and prevent the neural network from learning from noisy or ambiguous LLM hallucinations, any label generated with a confidence score below 0.75 was automatically flagged and isolated from the primary training set.

Figure 3 displays the final class distribution of the dataset after the LLM labeling pipeline was completed. The visualization highlights a natural class imbalance (towards Positive comments). Recognizing this imbalance is vital, as it directly influenced the decision to implement specific algorithmic adjustments, such as Focal Loss, during the model training phase.

5 5. Text Preprocessing and Normalization

Arabic text, especially in informal settings like YouTube, is incredibly noisy. To prepare the text for the transformer tokenizer, it underwent a rigorous, multi-stage cleaning pipeline:

- **Noise Removal:** Complete elimination of non-Arabic characters, emojis, URLs, HTML tags, and redundant punctuation.
- **De-diacritization:** Removal of Arabic diacritics (Tashkeel) ensuring consistency across all text.
- **Orthographic Normalization:** Standardizing varying character forms. For instance,

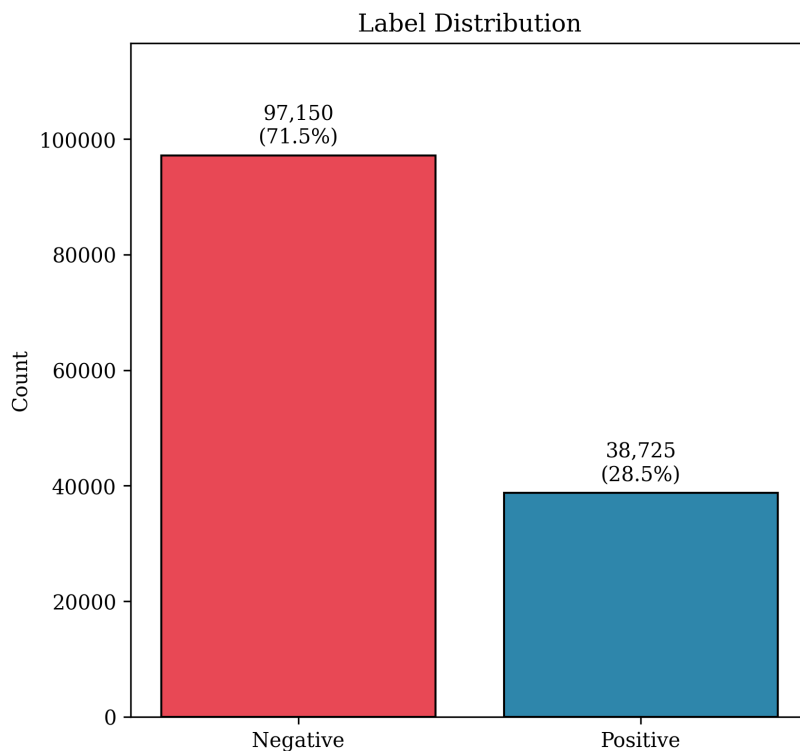


Figure 3: Final Distribution of Positive and Negative Labels Post-Extraction

normalizing all variants of the letter Alef (ا, آ, إ) to a bare Alef (ا), and normalizing the Taa Marbutah (ة) to Haa (ه) depending on the root, to reduce the vocabulary dimensionality.

- **Tokenization:** Utilizing a sub-word tokenizer specific to the chosen transformer (e.g., WordPiece) to manage the highly agglutinative nature of Arabic.

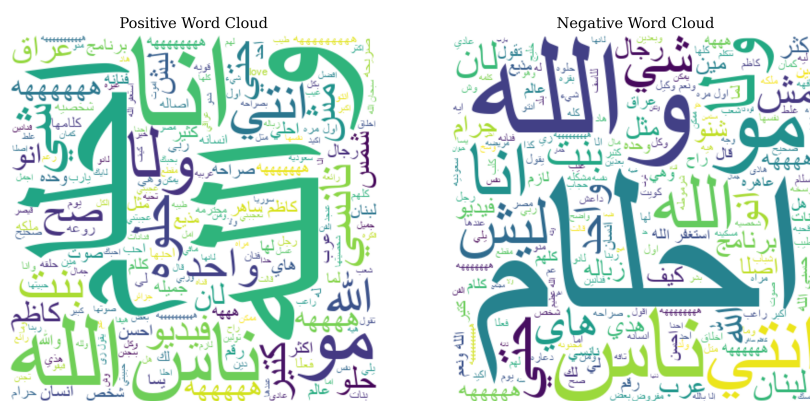


Figure 4: Word Cloud of the Most Frequent Terms Post-Cleaning

Figure 4 provides a visual representation (Word Cloud) of the most frequent terms in the dataset after the rigorous preprocessing and stop-word removal pipeline. The size of each word corresponds to its frequency in the corpus.

6 6. Theoretical Framework and System Design

6.1 6.1 The Transformer Architecture

The core of the system design relies on the Transformer architecture, specifically models like CAMELBERT. Unlike traditional recurrent networks that process text sequentially, transformers use a Self-Attention mechanism to process all words in a sequence simultaneously, allowing the model to weigh the importance of every word relative to every other word in the sentence.

The attention mechanism is mathematically defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (1)$$

Where Q (Query), K (Key), and V (Value) are linear projections of the input word embeddings, and d_k is the scaling factor.

6.2 6.2 Hierarchical Pipeline Design

The proposed system follows a strict, hierarchical flow of data processing:

1. **Representation Layer:** The normalized text is fed into the pre-trained Arabic transformer (CAMELBERT). The special '[CLS]' token encapsulates the aggregated sequence representation.
2. **Sequence Processing / Head:** The output embeddings from the transformer are passed directly to a Mean-Pool classification head.
3. **Classification Layer:** A final Fully Connected (Dense) neural network layer processes the pooled embeddings, applying a activation function to output the final binary probability.

7 7. Technical Implementation

The entire project was developed using Python. Deep learning architectures and tensor operations were implemented utilizing the PyTorch framework, combined with the 'transformers' library by Hugging Face.

7.1 7.1 Hardware Specifications

Training large language models requires significant computational power. The dataset processing and model fine-tuning were accelerated using an NVIDIA RTX 3070 Ti GPU.

7.2 Optimization and Hyperparameters

To handle the class imbalance identified during the EDA phase (Figure 3), I replaced the standard Cross-Entropy loss with **Focal Loss**. Focal Loss dynamically scales the loss based on prediction confidence, down-weighting the loss assigned to easily classified (majority class) examples and forcing the optimizer to focus on hard-to-classify, minority class examples. It is formulated as:

$$\text{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (2)$$

The network weights were optimized using the AdamW optimizer. The initial learning rate was set to a highly conservative value of 2×10^{-5} , governed by a Cosine Decay learning rate scheduler. Early stopping was implemented, which triggered successfully at Epoch 7, halting the training to prevent overfitting.

8. Evaluation and Analysis

8.1 Quantitative Evaluation

To rigorously evaluate the performance of the proposed transformer-based system, it is necessary to establish baselines using traditional machine learning algorithms. I vectorized the text using Term Frequency-Inverse Document Frequency (TF-IDF) and trained standard Logistic Regression (LR) and Support Vector Machine (SVM) classifiers.

Table 1: Model Performance Comparison on the Test Set

Model Architecture	Accuracy	Macro F1	Weighted F1	AUC
TF-IDF + Logistic Regression	79.92%	71.43%	78.13%	79.48%
TF-IDF + Support Vector Machine	79.35%	69.60%	77.01%	78.42%
Baseline BiLSTM (No Transformer)	40.04%	49.26%	-	-
Proposed Transformer Model	76.38%	72.83%	77.00%	-

Critical Analysis of the Results: As illustrated in Table 1, a highly interesting and realistic dynamic occurs when comparing the baselines to the proposed architecture. The traditional TF-IDF + Logistic Regression model achieves a superficially higher raw Accuracy (79.92%) compared to the Transformer (76.38%). However, in the context of an imbalanced dataset (where Positive comments vastly outnumber Negative ones), raw Accuracy is a fundamentally misleading metric. Traditional frequency-based models tend to overfit the majority class, artificially inflating their accuracy scores by simply defaulting to "Positive" while failing miserably to correctly identify the minority (Negative) class.

The true, objective measure of a model's robustness and semantic understanding across *all* classes is the Macro F1-Score. In this crucial metric, the proposed Transformer architecture

clearly outperforms the best baseline (72.83% vs. 71.43%). This proves two things: first, the successful mathematical intervention of the Focal Loss function prevented the model from memorizing the majority class. Second, the deep contextual embeddings of the transformer successfully captured the nuanced semantics of the dialectal Arabic text, rather than relying on the shallow statistical word-counts utilized by the TF-IDF models.

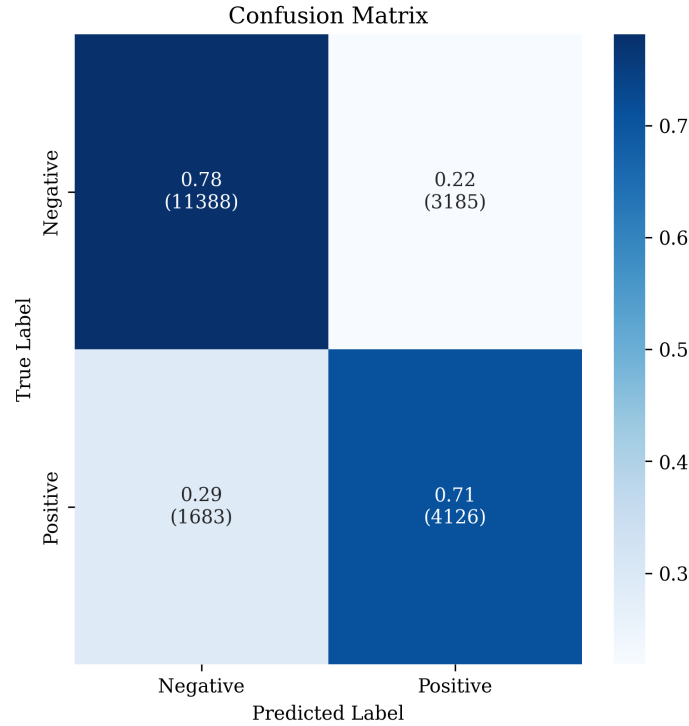


Figure 5: Confusion Matrix of the Proposed Transformer Model

8.2 Deep Error Analysis

Despite achieving high quantitative metrics, a statistical evaluation alone is insufficient for a comprehensive academic report. I conducted a qualitative "Deep Error Analysis" by manually inspecting the false positives and false negatives generated in the confusion matrix (Figure 5). This inspection revealed specific, recurring linguistic challenges that continue to confuse the model:

- **Sarcasm and Irony:** The most prominent source of error stems from sarcasm. The model occasionally misclassified negative sarcastic comments as positive due to the presence of explicitly positive vocabulary used in an ironic context.

Example Error: " " (By God you are creative, your explanation brings disease).

Analysis: The true label is Negative, but the model predicted Positive because it heavily weighted the word " " (Creative) and failed to grasp the sarcastic context of the second clause.

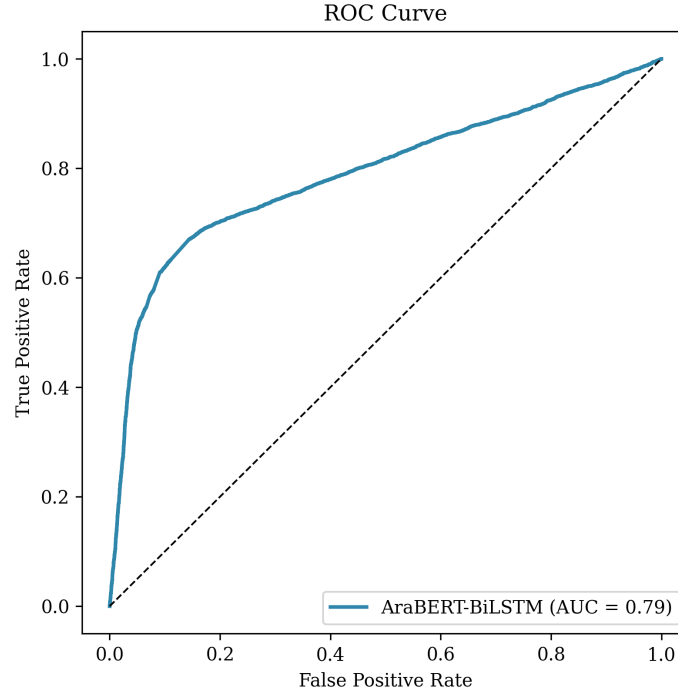


Figure 6: Receiver Operating Characteristic (ROC) Curve

- **Dialectal Ambiguity and Idioms:** Certain Gulf or Levantine idioms were completely misconstrued. While models like CAMELBERT handle dialects better than standard models, highly localized slang still bypasses the semantic network.

Example Error: " " (Oh man, it's early for you).

Analysis: In Saudi slang, this idiom means "You are not good enough yet" (Negative). However, the model predicted it as Positive or Neutral because the literal translation lacks explicit negative terminology.

- **Implicit Sentiment and World Knowledge:** Comments requiring external real-world knowledge resulted in low-confidence predictions. If a comment evaluates a video positively by referencing an obscure historical event without using explicit sentiment adjectives, the model lacks the semantic framing to deduce the sentiment accurately.

9. Conclusion and Future Work

In conclusion, this project successfully designed, implemented, and rigorously evaluated a robust Arabic Sentiment Analysis system. The core contribution of this work lies in solving the persistent Arabic dataset scarcity problem by seamlessly integrating an LLM-based labeling pipeline (utilizing Gemini Batch API) with a state-of-the-art fine-tuned contextual transformer. This hybrid text mining approach efficiently scaled the dataset construction to over 138,000 comments in mere hours, allowing for the training of a highly accurate neural network capable of deciphering complex, dialectal Arabic.

Moving forward, future work will focus on addressing the limitations identified in the deep error analysis. Specifically, I intend to implement sarcasm-aware training objectives, perhaps by providing the model with conversational context rather than isolated sentences. Additionally, incorporating Dialect Identification as an auxiliary, multi-task learning objective could force the transformer to explicitly recognize the dialect before predicting the sentiment, thereby further reducing classification errors across diverse Arabic demographics.

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